Capacity-feasible CVRP clustering on an analog neutral-atom register (Track A)

Pasqal challenge — team DuQuantum — repository ‘qhack‘

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1 1. Model

Problem (brief §2). Customers $i = 1..N$ with demands d_i, K identical vehicles of capacity Q, Euclidean costs. Cluster-first / route-second: the quantum part decides *which* customers share a vehicle; each cluster is then routed exactly (Held–Karp). Decision variables $x_{ik} \in \{0, 1\}$ (customer i on vehicle k), one-hot per customer, $n = NK$ atoms: **8 / 30 / 60** on S / M / L, no slack. Every derivation is in `docs/DERIVATIONS.md` (§1–§4) and `docs/MODEL.md`; this report states results and decisions.

Objective. Total pairwise distance inside each cluster, $f = \sum_k \sum_{i<j} d_{ij} x_{ik} x_{jk}$ — additive over clusters, so the exact set-partition DP (`cvrp/exact.py`) certifies the *model* optimum on every tier.

Constraints and what the device sees (DERIVATIONS §2–§4). At $\Omega = 0$, Eq. (1) assigns a bitstring the cost $E(x) = -\delta_f \sum_i x_i + \sum_{i<j} C_6 r_{ij}^{-6} x_i x_j$: one *uniform* linear coefficient (laser) and *positive* couplings (positions). Exactly-one [=]: squared penalty $A \sum_i (1 - \sum_k x_{ik})^2 \to h = -A$ on every variable (uniform (OK)), $J = 2A$ on each customer’s K-clique (positive (OK)). Capacity [≤]: unbalanced penalization $\sum_k [-\lambda_1 (Q - L_k) + \lambda_2 (Q - L_k)^2]$ (Montañez-Barrera et al. 2024), expanded; its per-customer linear part $c_i \sum_k x_{ik}$ is a *constant* on one-hot strings (the one-hot identity), so it is dropped — λ_1 disappears and what remains is the coupling $+2\lambda_2 d_i d_j$ on same-vehicle pairs, i.e. a **load-balance term** $\lambda_2 \sum_k L_k^2$. It is soft: for $K = 2$ the balance optimum is capacity-feasible whenever a feasible split exists, for $K \geq 3$ only likely, so we score feasible decodes only and report the feasibility rate (Pitfall 9.1). Weights: $A = \max_i \sum_{j \neq i} (d_{ij} + 2\lambda_2 d_i d_j) + 1$ (provable: every argmin is one-hot), $\lambda_2 = 0.15\, R/(2Q)$ from a 40-seed brute-force scan. Brute-force checker on tier S: argmin feasible and optimal on 5/5 shipped instances; A survives one halving and fails at ¼; λ_2 is not binding on these instances (`out/gate1/`).

2 2. Variable accounting (brief Derive 4.1)

Encoding	count	S	M	L	fits 80?
assignment variables only	NK	8	30	60	yes
+ binary slack (textbook)	$NK + K\lceil \log_2(Q{+}1) \rceil$	14	45	80	yes (L = cap)
ours (`onehot-ub`)	NK	**8**	**30**	**60**	yes

(`scripts/variable_count.py`; the textbook rows are implemented and checked, DERIVATIONS §3.) Saving: 6 / 15 / 20 atoms, and no atom whose readout is discarded; the capacity term adds no coupling pairs beyond the objective’s.

3 3. Embedding (DERIVATIONS §5)

Fidelity formula reported everywhere: $\|U/\alpha - J\|_F / \|J\|_F$ over off-diagonals, with the worst-pair error next to it. The generic least-squares embedder cannot realise the one-hot rook’s graph: the K copies of a customer sit 5 µm apart, so the “zero” cross-vehicle couplings come out as large as the wanted ones (rel. Frobenius 0.95; on S_seed04 the register’s own ground state was infeasible). We therefore ask for less: on one-hot strings each customer pair contributes exactly one coupling, so only *same-vehicle minus cross-vehicle* must equal αw_{ij} (identity 5.7b). Each customer becomes a rigid 5 µm K-gon (clique pinned at the minimum separation, $\alpha = C_6/5^6/2A$); centres and orientations are fitted so the device energies of all one-hot strings match αE_{QUBO}, with explicit single-flip defect gaps and a fitted δ_f (a constant on the subspace). Registers are cached and committed (`embeddings/`). Diagnostics per run: full-matrix fidelity, subspace fidelity, and the realised-register check (device argmin feasible? equal to the model optimum? defect gap). On tier S: argmin = model optimum on 5/5 instances.

Where it breaks (tier M, $K = 3$). Triangles 5 µm on a side cannot be far from each other’s neighbours: the subspace-fit residual is 38 (S: 0.15–0.32), defects fall below their parents and the model optimum ranks ~10 000th among one-hot strings; opening the polygons lowers the energy scale without fixing the ordering (DERIVATIONS §5.7, STATUS). This is the brief’s §4 warning in its sharpest form, and it is a property of one-hot encodings on a 2-D $1/r^6$ register.

4 4. Pulse (DERIVATIONS §6)

Three-point Ω ramp (15 % / 70 % / 15 %) and a linear δ sweep from $\delta_0 = -\min(\max(|\delta_f|, 2\Omega_{\max}), \delta_{\max})$ to δ_f while Ω is on its plateau; every sequence is built with Pulser on `AnalogDevice` and validated (`to_abstract_repr()`, limits read from the device object). Scan on three tier-S instances, scored by P(reach), best gap and feasibility (never mean cost): longer is monotonically better; $\Omega = 0.75\,\Omega_{\max}$ beats full and half power; $\delta_f \times \{0.8, 1, 1.25\}$ peaks at the fitted value. **Chosen: 5800 ns,** $0.75\,\Omega_{\max}$**, fitted** δ_f (35–38 rad/µs on S, inside ±125.66). Emulators: pulser-simulation (qutip) for every headline row, emu-sv as a cross-check, and an in-house exact state-vector integrator built from Pulser's own samples for scans (agrees with qutip to sampling noise, TV ≈ 0.05 at 500 shots); emu-mps for M/L with the bond dimension recorded.

5 5. Results (brief Table 3; `results.csv`, `python -m cvrp.results`)

instance	method	feas.	model gap	CVRP gap	P(reach C*)	P(gap≤5%)	fidelity	backend
S_seed00	ours	1.00	0.0 %	0.0 %	0.94	0.94	0.26	qutip
S_seed00	ours, generic emb.	0.22	0.0 %	0.0 %	0.13	0.13	0.95	qutip
S_seed00	qubo-solver quantum	1.00	0.0 %	0.0 %	0.78	0.78	0.12	qs-quantum
S_seed00	qubo-solver tabu	1.00	0.0 %	0.0 %	1.00	1.00	undef.	qs-classical
S_seed00	random sampler	0.04	0.0 %	0.0 %	0.01	0.01	0.95	random
S_seed01	ours	0.95	0.0 %	0.0 %	0.95	0.95	0.21	qutip
S_seed01	ours, generic emb.	0.09	0.0 %	0.0 %	0.07	0.07	0.94	qutip
S_seed01	qubo-solver quantum	0.80	0.0 %	0.0 %	0.78	0.78	0.16	qs-quantum
S_seed01	qubo-solver tabu	1.00	0.0 %	0.0 %	1.00	1.00	undef.	qs-classical
S_seed01	random sampler	0.02	0.0 %	0.0 %	0.01	0.01	0.94	random
S_seed02	ours	1.00	0.0 %	0.0 %	0.99	0.99	0.31	qutip
S_seed02	qubo-solver quantum	0.95	0.0 %	0.0 %	0.18	0.18	0.21	qs-quantum
S_seed02	qubo-solver tabu	1.00	0.0 %	0.0 %	1.00	1.00	undef.	qs-classical
S_seed02	random sampler	0.04	0.0 %	0.0 %	0.01	0.01	0.31	random
S_seed03	ours	1.00	0.0 %	15.6 %	0.01	0.96	0.29	qutip
S_seed03	qubo-solver quantum	0.99	0.0 %	15.6 %	0.18	0.79	0.14	qs-quantum
S_seed03	qubo-solver tabu	1.00	0.0 %	15.6 %	0.00	1.00	undef.	qs-classical
S_seed03	random sampler	0.05	0.0 %	15.6 %	0.01	0.01	0.29	random
S_seed04	ours	1.00	0.0 %	21.7 %	0.02	0.89	0.32	qutip
S_seed04	ours, generic emb.	0.06	0.0 %	21.7 %	0.00	0.03	0.95	qutip
S_seed04	qubo-solver quantum	1.00	27.3 %	30.0 %	0.00	0.00	0.19	qs-quantum
S_seed04	qubo-solver tabu	1.00	0.0 %	21.7 %	0.00	1.00	undef.	qs-classical

instance	method	feas.	model gap	CVRP gap	P(reach C*)	P(gap≤5%)	fidelity	backend
S_seed04	random sampler	0.04	0.0 %	21.7 %	0.01	0.01	0.95	random

(generated by `python scripts/report_tables.py` from `results.csv`; raw shots only — repair rows are in results.csv under their own label; shots: 500 per emulator row, 2000 for the random sampler, 1 for a classical solver)

References are proven optima for every instance (exhaustive set-partition DP × exact TSP; SA and naive enumeration agree; CPLEX unavailable). Decomposition gap (model optimum routed exactly vs C^*):

instance	ρ	C* (proven)	model optimum routed exactly	decomposition gap
S_seed00	0.714	262.97	262.97	0.0 %
S_seed01	0.857	245.02	245.02	0.0 %
S_seed02	0.714	254.62	254.62	0.0 %
S_seed03	0.643	197.44	228.34	15.6 %
S_seed04	0.714	184.42	224.49	21.7 %

Reading: with the structure-aware register and the scanned pulse, the device returns the model-optimal clustering in 89–99 % of raw shots on all five tier-S instances (random sampler: 0.7–1 %; generic embedder: 3–13 %). Where P(reach C^*) is low with a 0 % model gap (S_seed03/04) the loss is the decomposition, not the device. qubo-solver's own pipeline (BLaDE + its drive shaper, 5940 ns) reaches 0.18–0.79 of the model optimum on four instances and fails on S_seed04 (27 % model gap); its classical tabu search finds every model optimum, and its default SA path does not (a single-start greedy descent at $T_0 = 5$ — we report a stated-parameter SA instead).

6 6. What did not work

See `docs/DID_NOT_WORK.md` (one sentence each): no PyPI/GitHub from the sandboxes (Windows job runner instead); CPLEX not installable; a typed number caught by review; qubo-solver's default SA; the emu-mps calibration job blocking the queue; cross-platform non-reproducibility of the embedding optimiser (fixed by committing registers); the one-hot rook's graph not embedding for $K = 3$ (tier M), which is the honest limit of this encoding on this device.

7 7. Use of AI

`docs/AI_USE.md`. Claude wrote the code, the plumbing and most of the prose from the team's decisions; the Gate-1 formulation (`docs/MODEL.md`) was AI-drafted at the team's request under time pressure and checked by brute force; the team must re-derive it before judging. Every number in this report is produced by a script from committed records.